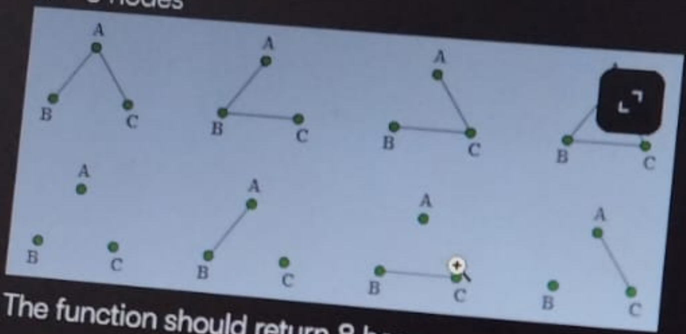
Drawing Edge

Given a number of nodes, how many ways can a graph be drawn?

A network consists of multiple nodes. Each pair of distinct nodes can either be connected by an edge or remain disconnected. A node cannot be connected to itself. Determine the total number of distinct configurations of such a network. Since the result can be very large, return it modulo (10^9+7).

Example 1

n = 3 nodes

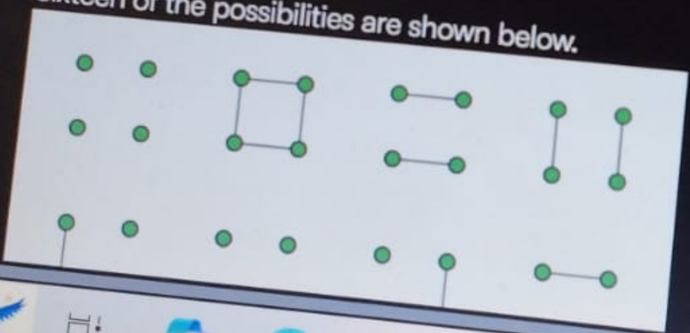


The function should return 8 because there are 8 possible configurations.

Example 2

n = 4 nodes

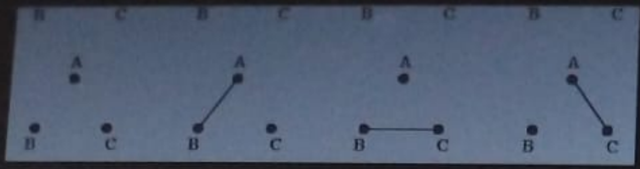
Sixteen of the possibilities are shown below.



Language C

```
19  #incluc
int draw
}
> int main(
```

Test Results

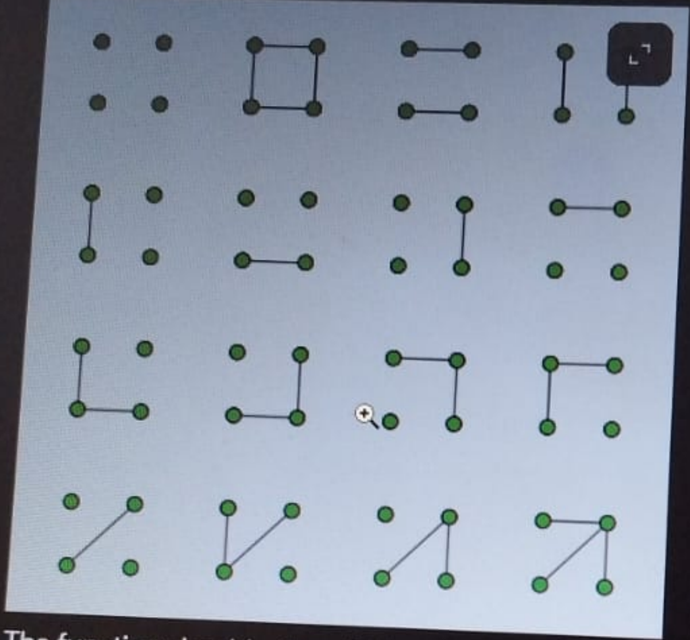


The function should return 8 because there are 8 possible configurations.

Example 2

n = 4 nodes

Sixteen of the possibilities are shown below.



The function should return 64 because there are 64 possible configurations.

Constraints

- $1 \le n \le 10^9$

Language C

```
#include <
19

int drawingEdg
}
> int main()
```

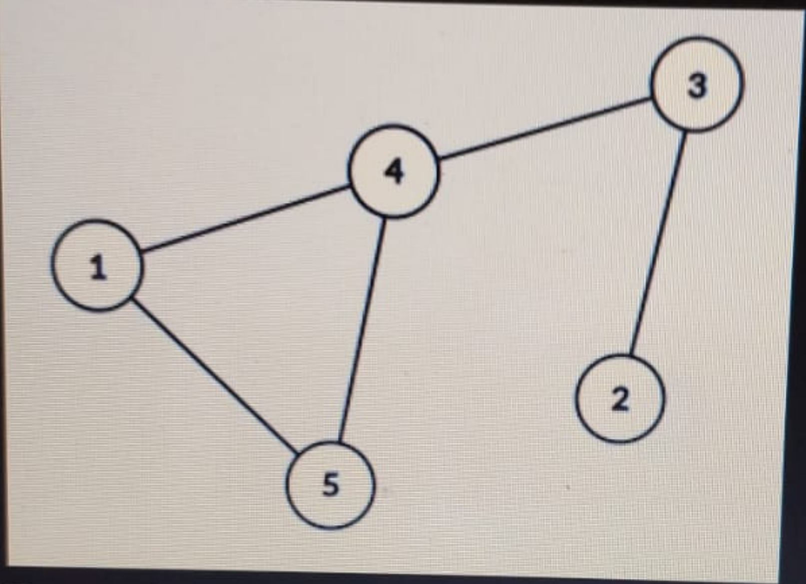
Test Results

2/300d80nq9pp/questions/4ss7enarja3

g_to = [5, 1, 4, 3, 2]

There are g_nodes = 5 nodes and M = 5 edges.
Connected pairs are elements of *g_from* and g_to,:
(4, 5), (5, 1), (1, 4), (4, 3), (3, 2).

The graph looks like this:



Traverse the graph first and store the order of traversal in array A. Determining the order of traversal is up to you. For this example, the optimal traversal is:

5 -> 4 -> 3 -> 2 -> 3 -> 4 -> 1
A = [5, 4, 3, 2, 3, 4, 1]

Array B, according to the described algorithm is:
B = [5, 4, 3, 2, 1]

This is the largest lexicographically possible array B.

Function Description
Complete the function *coolGraph* in the editor

Language

```
> #in
/*
 *
 *
 * T
 * T
 */

/*
 * F
 *
 * 1.
 * 2.
 * 3.
 *
 */

/*
 * To
 *
 *
 *
 * For
 * int*
 *
 *
 *
 *
 *
```

Test Results

est-v2/300d80nq9pp/questions/4ss7enarja3

▶ Input Format For Custom Testing

▼ Sample Case 0

Sample Input For Custom Testing

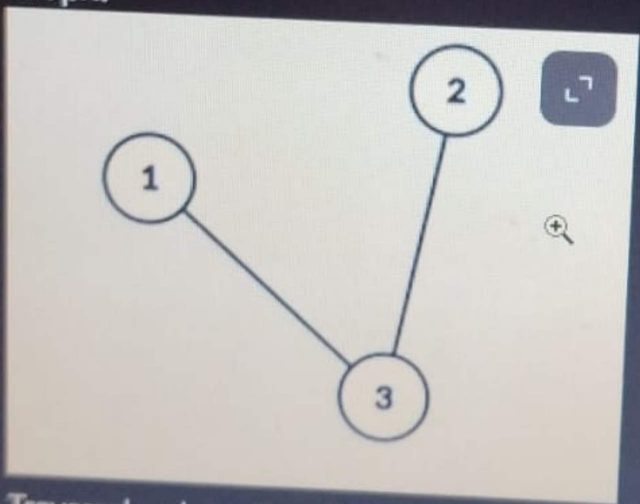
```
STDIN     Function
-----     --------
3 2       g_nodes = 3, M = 2
3 1       g_from[] = [3, 3], g_to = [1,
2]
3 2
```

Sample Output

```
3 2 1
```

Explanation

Graph:



Traversal order = 3 -> 2 -> 3 -> 1. Array A = [3, 2, 3, 1]

Array B according to the given algorithm: Array B = [3, 2, 1]

▶ Sample Case 1

Language

Test Results

n/test-v2/300d80nq9pp/questions/4ss7enarja3

55 sec

This is the largest lexicographically possible array B.

Function Description

Complete the function *coolGraph* in the editor below. The function must return an array representing array B.

coolGraph has the following parameters:

int g_nodes: the number of nodes

int g_from[M]: one end of each connected pair of nodes

int g_to[M]: the other end of each connected pair of nodes

Constraints

- $1 \le g\_nodes \le 10^5$
- $1 \le M \le 2 * 10^5$
- $1 \le g\_from[i] \le 2 * 10^5 \ (1 \le i \le M)$
- $1 \le g\_to[i] \le 2 * 10^5 \ (1 \le i \le M)$
- $g\_from[i] \ne g\_to[i] \ (1 \le i \le M)$

▶ Input Format For Custom Testing

▼ Sample Case 0

Sample Input For Custom Testing

```
STDIN     Function
-----     --------
3 2       g_nodes = 3, M = 2
3 1       g_from[] = [3, 3], g_to = [1,
2]
3 2
```

Lang

Test Res

MathWorks 1 hr 21 min

(Math Question) Divisibility by 7 or 9

What is the probability that a randomly chosen integer from 1 to 100 is divisible by 7 or 9?

(Math Question) Divisibility by 7 or 9

Pick ONE option

- 0.15
- 0.20
- 0.24
- 0.30

Clear Selection

MathWorks

1 hr 21 min

(Math Question) Arranging boys and girls

In how many ways can 4 boys and 3 girls stand in a line if they must alternate?

(Math Question) Arranging boys and girls

Pick ONE option

- 144
- 288
- 504
- 720

Clear Selection

hackerrank.com/test-v2/g2m56ifciro/questions/kc74id62fc

MathWorks

1 hr 21 min

(Math Question) Playing Cards

A standard deck of 52 playing cards is shuffled thoroughly. What is the probability of drawing a red card followed by a black card, without replacement?

(Math Question) Playing Cards

Pick ONE option

- 1/2
- 25/102
- 1/4
- 13/51

Clear Selection

hackerrank.com/test-v2/g2m56ifciro/questions/kc74id62fc

Search

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hackerrank.com/test-v2/g2m56ifciro/questions/2a5rc0gs6s4

MathWorks

1 hr 21 min

(Math Question) Committee of 5 members

A committee of 5 people is to be formed from a group of 6 men and 4 women. In how many ways can the committee be formed if it must have at least 2 women?

(Math Question) Committee of 5 members

Pick ONE option

- 145
- 215
- 186
- 340

Clear Selection

hackerrank.com/test-v2/g2m56ifciro/questions/2a5rc0gs6s4

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ank.com/test-v2/g2m56ifciro/questions/2sec8rrokpp

1 hr 21 min

(Prog. Concept Question) TC connection

Which of the following always use persistent TCP connection?

Pick ONE or MORE options

- [ ] HTTP
- [ ] FTP
- [ ] SMTP
- [ ] All of above

Clear Selection

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hackerrank.com/test-v2/g2m56ifciro/questions/1kqltqfgbq2

MathWorks 1 hr 21 min

(Math Question) Remainder on dividing 2^2021

What is the remainder when 2^{2021} is divided by 17?

Pick ONE option

- 1
- 2
- 7
- 15

Clear Selection

(Math Question) Remainder on dividing 2^2021

hackerrank.com/test-v2/g2m56ifciro/questions/1kqltqfgbq2

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m/test-v2/g2m56ifciro/questions/8gjb59sp9to

0 min

(Prog. Concept Question) Page Table

A computer system implements a 36 bit virtual address. Page size is 4KB and size of physical memory is 30 bits. What is the approximate size of page table in MB?

Pick ONE option

- 32 MB
- 48 MB
- 42 MB
- 16 MB

Clear Selection

iro/questions/8gjb59sp9to

Search

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n/test-v2/g2m56ifciro/questions/70rircfh8j7

min

(Prog. Concept Question) Deadlock

Which one of the following is a way to avoid deadlock?

Pick ONE option

- Fragmentation
- Thrashing
- Spooling
- None of the above

Clear Selection

/questions/70rircfh8j7

Search

1 hr 19 min

(C Question) printf Statement

What does the printf statement prints?

```
#include<stdio.h>

int main()

{

   int i=10;

   int k=0;

   k==1>0?i++:2;

   printf("%d", i);

   return 0;

}
```

Pick ONE option

- 10
- 11
- 12
- None of these

Clear Selection

rank.com/test-v2/g2m56ifciro/questions/g4mp7bgfjma

1 hr 19 min

(C Question) Operator

Predict the output of the given program:

```
#include <stdio.h>

#define test(a, b) a##b

int main()

{

printf("%d ", test(34, 45));

}
```

Pick ONE option

- 3445
- 345
- 0
- 1
- Compile Error

Clear Selection

(Prog. Concept Question) Binary Tree

In which of the following options, we cannot construct a unique Binary tree?

Pick ONE or MORE options

- [ ] Postorder and Preorder.
- [ ] Inorder and Level-order.
- [ ] Postorder and Level-order.

Clear Selection

(C Question) Operator sequence II

Predict the output of following C program:

```
#include<stdio.h>

void main()

{

int i=6;

printf("%d %d %d %d
%d\n",!i,++i,i+3,i--,i++) ;

}
```

Pick ONE option

- 0 7 10 7 6
- 1 7 10 7 6
- 0 7 9 7 6
- None of these

Clear Selection

(C Question) Loop

How many times 'MATLAB' will be printed in the following C code?

```
#include <stdio.h>

int main()

{

    int i = 1024;

    for (; i; i >>= 1)

        printf("MATLAB");

    return 0;

}
```

Pick ONE option

- 10
- 11
- 9
- Infinite

Clear Selection

(C++ Question) Simple Output Problem

What is the output for the following?

```
#include<iostream>


int main()

{

  int i=10;

  std::cout<<"%d"<<i;

}
```

Pick ONE option

- 10
- Compiler error
- Runtime error
- None of the above

Clear Selection

hackerrank.com/test-v2/g2m56ifciro/questions/g68mrod1dqt

thWorks ⊙ 1 hr 17 min

(C++) Find the output of the following code.

```cpp
#include <bits/stdc++.h>

class MathWorks {
public:
   virtual void print() {
      std::cout << "MathWorks" << std::endl;
   }
};

class Employer : public MathWorks {
public:
   void print() override {
      std::cout << "Employer" << std::endl;
   }
};

class Employee : public Employer {
public:
```

Pick ONE option

- ○ MathWorks
- ○ Employer
- ○ Employee
- ○ Compilation Error

Clear Selection

(C Question) Pointers I

Predict the output of following program:

```
#include<stdio.h>
void main()
{
int a[]={1,2,4,6,8};
int* p[]={a,a+1,a+2,a+3,a+4};
int** p1=p;
int* p2=*(p+2);
printf("%u %u %u\n",*++p2,++*p2++,*++*++p1);
}
```

Pick ONE option

- 674
- 975
- 985
- 864

Clear Selection

ank.com/test-v2/g2m56ifciro/questions/a2qginikmb2

1 hr 16 min

(C/C++) Find the output of t following code.

```
#include <stdio.h>


int main(){
   int x = 5;
   int * const ptr = &x;
   ++(*ptr);
   printf("%d", x);
   return 0;
}
```

Pick ONE option

- 6
- 5
- Compilation Error
- Runtime Error

Clear Selection

```cpp
    }
};

class Employee : public Employer {
public:
    void print() override {
        std::cout << "Employee" << std::endl;
    }
};


int main() {

    MathWorks* ptrA = new Employee;
    Employer* ptrB = dynamic_cast<Employer*>(ptrA);
    ptrB->print();
    return 0;
}
```

Pick ONE option

- MathWorks
- Employer
- Employee
- Compilation Error

Clear Selection

Search

```
int num;
public:
void set(int n) {
num = n;
}
void disp() {
cout<<"Stored Number : "<<num;
}
};
};
int main() {
A :: B obj;
obj.set(5);
obj.disp();
return 0;
}
```

Pick ONE option

- Stored Number : 5
- Runtime Error
- Compilation Error

Clear Selection

(C++) Find the output of the following question.

```
#include <iostream>

int main() {

    int x = 5;

    int y = (x++) + (++x) + (x++);

    std::cout << y << std::endl;

    return 0;

}
```

Pick ONE option

- ○ 17
- ○ 16
- ○ 15
- ○ Depends on the compiler

Clear Selection

1 hr 16 min

(C++) Find the output of the following question.

```
#include<iostream>
using namespace std;
class A {
  public:
  class B {
    private:
    int num;
    public:
    void set(int n) {
      num = n;
    }
    void disp() {
      cout<<"Stored Number : "<<num;
    }
```

Pick ONE option

- Stored Number : 5
- Runtime Error
- Compilation Error

Clear Selection

Search

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hackerrank.com/test-v2/g2m56ifciro/questions/6rb1krr306j

MathWorks 1 hr 24 min

Sample Output

```
0
1
1
0
0
```

Explanation

The number of divisors and multiples of 5, 3, and 7 are 0.

The number of divisors and multiples of 2, 4 are 1

▼ **Sample Case 1**

Sample Input For Custom Testing

```
STDIN     FUNCTION
-----     --------
4      →  arr size n = 4
2      →  arr = [2, 4, 8, 16]
4
8
16
```

Sample Output

```
3
3
3
3
```

Explanation

Every element in the array is either a divisor or a multiple of another element.

cloudy Search

hackerrank.com/test-v2/g2m56ifciro/questions/6rb1krr306j

rks

1 hr 24 min

Language C

```c
#include <assert.h>...

/*
 * Complete the 'getCount' function below.
 *
 * The function is expected to return an INTEGER_ARRAY.
 * The function accepts INTEGER_ARRAY arr as parameter.
 */

/*
 * To return the integer array from the function, you should:
 *     - Store the size of the array to be returned in the result_count variable
 *     - Allocate the array statically or dynamically
 *
 * For example,
 * int* return_integer_array_using_static_allocation(int* result_count) {
 *     *result_count = 5;
 *
 *     static int a[5] = {1, 2, 3, 4, 5};
 *
 *     return a;
 * }
 *
 * int* return_integer_array_using_dynamic_allocation(int* result_count) {
 *     *result_count = 5;
 *
 *     int *a = malloc(5 * sizeof(int));
 *
 *     for (int i = 0; i < 5; i++) {
```

Test Results

Search

- $1 \le arr[i] \le 10^5$

▼ Input Format For Custom Testing

The first line contains an integer, *n*, denoting the number of elements in *arr*.
Each line *i* of the *n* subsequent lines (where $0 \le i < n$) contains an integer describing *arr[i]*.

▼ Sample Case 0

Sample Input For Custom Testing

```
STDIN     FUNCTION
-----     --------
5    →    arr size n = 5
5    →    arr = [5, 2, 4, 3, 7]
2
4
3
7
```

Sample Output

```
0
1
1
0
0
```

Explanation

The number of divisors and multiples of 5, 3, and 7 are 0.
The number of divisors and multiples of 2, 4 are 1

▶ Sample Case 1

Divisors and Multiples

Given an array *arr* of length *n*, for each index *i* from 1 to *n*, count how many indices *j* (where *j* is not equal to *i*) satisfy either of the following conditions:

- *arr[j]* is a divisor of *arr[i]*
- *arr[j]* is a multiple of *arr[i]*

Definitions:

- *x* is a divisor of *y* if *y* is divisible by *x* (y % x = 0).
- *x* is a multiple of *y* if *x* is divisible by *y* (x % y = 0).

Return an array where the i^{th} element represents the count for the i^{th} element of the input array.

Example

arr = [1, 3, 4, 2, 6]

i	*arr[i]*	Divisors	Multiples	Count
0	1	0	4	0 + 4 = 0
1	3	1	1	1 + 1 = 2
2	4	2	0	2 + 0 = 0
3	2	1	2	1 + 2 = 3
4	6	3	0	3 + 0 = 3

For *arr[0]* = 1, there are no divisors in the array, and the other 4 elements are multiples of 1. Return the counts as an array, [4, 2, 2, 3, 3].

Function Description

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Works 1 hr 24 min

Language C

```c
 *      - Allocate the array statically or dynamically
 *
 * For example,
 * int* return_integer_array_using_static_allocation(int* result_count) {
 *     *result_count = 5;
 *
 *     static int a[5] = {1, 2, 3, 4, 5};
 *
 *     return a;
 * }
 *
 * int* return_integer_array_using_dynamic_allocation(int* result_count) {
 *     *result_count = 5;
 *
 *     int *a = malloc(5 * sizeof(int));
 *
 *     for (int i = 0; i < 5; i++) {
 *         *(a + i) = i + 1;
 *     }
 *
 *     return a;
 * }
 *
 */
int* getCount(int arr_count, int* arr, int* result_count) {

}

> int main()...
```

Test Results

Search

Task Completion

Two interns at HackerRank are assigned to complete a total of *n* tasks. Each task can be completed by either intern. The first intern earns *reward_1[i]* points for finishing the ith task, while the second intern earns *reward_2[i]* points for the same task.

To maximize the total reward points for both interns, determine the highest possible combined reward points if the first intern is required to complete *k* tasks, and the second intern completes the remaining tasks.

Note: The *k* tasks for the first intern can be any of the *n* tasks.

Example

Consider *n = 5, reward_1 = [5, 4, 3, 2, 1], reward_2 = [1, 2, 3, 4, 5]* and *k = 3*.

Intern 1 completes 3 tasks, while intern 2 completes the other 2. To maximize the points, intern 1 takes the first 3 tasks, and intern 2 takes the last 2 tasks. The total reward points are 5 + 4 + 3 (from intern 1) + 4 + 5 (from intern 2) = 21, which is the maximum possible. So, the answer is 21.

Function Description:

Complete the function *getMaximumRewardPoints* in the editor with the following parameters:

int k: the number of tasks that have to be completed by intern 1

int reward_1[n]: the reward points earned by intern 1 for each task

int reward_2[n]: the reward points earned by intern 2 for each task

Returns

int: the maximum possible combined reward points when intern 1 completes exactly *k* tasks

Constraints:

```c
#include <assert.h>...

/*
 * Complete the 'getMaximumRewardPoints' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts following parameters:
 *  1. INTEGER k
 *  2. INTEGER_ARRAY reward_1
 *  3. INTEGER_ARRAY reward_2
 */

int getMaximumRewardPoints(int k, int reward_1_count, int* reward_1, int reward_2_count, int* reward_2) {

}

int main()...
```

Language C

Layout

Save & Proceed

```c
#include <assert.h>...

/*
 * Complete the 'getMaximumRewardPoints' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts following parameters:
 *  1. INTEGER k
 *  2. INTEGER_ARRAY reward_1
 *  3. INTEGER_ARRAY reward_2
 */

int getMaximumRewardPoints(int k, int reward_1_count, int* reward_1, int reward_2_count, int* reward_2) {

}

int main()...
```

Ln 19, Col 1 Autocomplete Spaces: 4 Mode:

Run

Test Results

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hackerrank.com/test-v2/g2m56ifciro/questions/6rb1krr306j

MathWorks ⏱ 1 hr 24 min

2	4	2	0	2 + 0 = 0
3	2	1	2	1 + 2 = 3
4	6	3	0	3 + 0 = 3

For *arr[0]* = 1, there are no divisors in the array, and the other 4 elements are multiples of 1. Return the counts as an array, [4, 2, 2, 3, 3].

Function Description

Complete the function *getCount* in the editor with the following parameter(s):

int arr[n]: an array of integers

Returns

int[n]: the answers at each index

Constraints

- $1 \le n \le 10^5$
- $1 \le arr[i] \le 10^5$

▼ Input Format For Custom Testing

The first line contains an integer, *n*, denoting the number of elements in *arr*.
Each line *i* of the *n* subsequent lines (where $0 \le i < n$) contains an integer describing *arr[i]*.

▼ Sample Case 0

Sample Input For Custom Testing

```
STDIN    FUNCTION
-----    --------
5     →  arr size n = 5
```

Intern 1 completes 3 tasks, while intern 2 completes the other 2. To maximize the points, intern 1 takes the first 3 tasks, and intern 2 takes the last 2 tasks. The total reward points are 5 + 4 + 3 (from intern 1) + 4 + 5 (from intern 2) = 21, which is the maximum possible. So, the answer is 21.

Function Description:

Complete the function *getMaximumRewardPoints* in the editor with the following parameters:

int k: the number of tasks that have to be completed by intern 1

int reward_1[n]: the reward points earned by intern 1 for each task

int reward_2[n]: the reward points earned by intern 2 for each task

Returns

int: the maximum possible combined reward points when intern 1 completes exactly *k* tasks

Constraints:

- $1 \le n \le 10^5$
- $0 \le k \le n$
- $1 \le reward\_1[i] \le 10^4$
- $1 \le reward\_2[i] \le 10^4$

▼ Input Format For Custom Testing

The first line contains an integer, *k*.
The next line contains an integer, *n*, the number of tasks.
The next *n* lines contain an integer, *reward_1[i]*.
The next line contains an integer, *n*, the number of tasks.
The next *n* lines contain an integer, *reward_2[i]*.

Language: C

```c
#include <assert.h>...

/*
 * Complete the 'getMaximumRewardPoints' function
 *
 * The function is expected to return an INTEGER.
 * The function accepts following parameters:
 *  1. INTEGER k
 *  2. INTEGER_ARRAY reward_1
 *  3. INTEGER_ARRAY reward_2
 */

int getMaximumRewardPoints(int k, int reward_1_c
reward_2_count, int* reward_2) {

}

int main()...
```

Ln 19, Col 1

^ Test Results

MathWorks 1 hr 26 min

The next line contains an integer, *n*, the number of tasks.
The next *n* lines contain an integer, *reward_1[i]*.
The next line contains an integer, *n*, the number of tasks.
The next *n* lines contain an integer, *reward_2[i]*.

▼ Sample Case 0

Sample Input For Custom Testing

```
STDIN          FUNCTION
-----          --------
3         →    k = 3
4         →    reward_1[] size, n = 4
1         →    reward_1 = [1, 2, 3, 2]
2
3
2
4         →    reward_2[] size, n = 4
1         →    reward_2 = [1, 2, 3, 2]
2
3
2
```

Sample Output

```
8
```

Explanation

Intern 1 has to complete 3 tasks, and intern 2 completes the remaining task. The reward points for each task are the same for both interns, so any task can be picked up by either intern. Total reward points = 1 + 2 + 3 + 2 = 8.

▶ Sample Case 1

Layout

Language C

```c
#include <assert.h>...

/*
 * Complete the 'getMaximumRewardPoints' funct
 *
 * The function is expected to return an INTE
 * The function accepts following parameters:
 *  1. INTEGER k
 *  2. INTEGER_ARRAY reward_1
 *  3. INTEGER_ARRAY reward_2
 */

int getMaximumRewardPoints(int k, int rewar
reward_2_count, int* reward_2) {

}

int main()...
```

Test Results